ChronoQA: Benchmarking Temporal Grounding in Long-Form Video via Timestamp-Anchored Multiple-Choice QA

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Abstract

The expanding use of large language models in intelligent workflows has triggered rapid advancement in Multimodal Large Language Models (MLLMs). Despite their success with short-form video, existing models still fall short when tasked with long-duration sequences. Temporal grounding, which involves localizing when an event occurs within a video, is a persistent weakness of video-language models (VLMs), in part because existing training data provides answer supervision but lacks temporal evidence supervision. Without it, artificial intelligence cannot reliably assist with tasks such as medical surgery review, security monitoring, or analyzing long-form video data. Yet, existing benchmarks evaluate answer correctness without verifying whether a model identifies the correct supporting temporal evidence. We introduce **ChronoQA**, a multiple-choice QA benchmark that pairs every correct answer with a timestamp interval, providing an explicit grounding target that other video QA datasets omit, and enabling evaluation of answer accuracy and temporal localization quality. We construct ChronoQA from 423 long-form YouTube videos (10-90 minutes each), yielding 897 timestamp-anchored multiple-choice QA pairs across five task categories: Action Understanding, Needle-in-a-Haystack, Narrative Understanding, Causal Understanding, and Ordering. Through multimodel blind filtering, we constrain text-only solvability to 17.8%, near the 20% random baseline, ensuring visual perception rather than linguistic bias and visual information leakage.

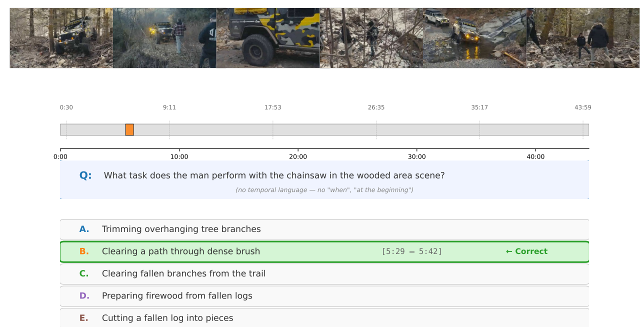

Figure 1: An illustration of a multiple-choice question samples from our temporal grounding benchmark, ChronoQA.

1 Introduction

While conventional video models are proficient at determining whether an event occurred, they often struggle to pinpoint its timing and duration. A student reviewing a recorded lecture must locate the exact moment a concept is introduced, understand how it is developed over the rest of the video, and identify the follow-up examples that make it concrete enough to apply.

Temporal grounding addresses this challenge by anchoring semantic understanding to a rigorous timeline, which is crucial for effectively addressing complex video understanding tasks [12]. This anchoring enables precise retrieval of specific moments via natural-language queries, causal linking of events separated by minutes, and structured summarization in which every claim traces back to a verifiable segment. Without robust temporal grounding, the outputs of video-language models remain ungrounded, lacking the evidence required for applications where temporal accuracy is non-negotiable. By enforcing a strict mapping between semantic content and the temporal axis, we enable a deeper, more structured evidence-based form of video reasoning necessary for deploying VLMs in medical, legal, and industrial auditing. However, without a benchmark that jointly evaluates answer correctness and temporal localization, there is no way to determine whether any of these capabilities have actually been achieved.

Existing video benchmarks provide questions and answers but lack explicit temporal grounding. They do not direct models to the specific moments where answers can be found. This leaves models unable to reliably locate key events or reason precisely about when things occur in a video. While some works propose temporal understanding, they remain clip-level discrimination tasks that neither require models to localize events in time nor reason about their precise temporal position within a video, leaving temporal localization entirely unaddressed [2]. Without the isolation of the temporal structure, models are left to rely on static visual cues or linguistic shortcuts. Meanwhile, traditional temporal grounding datasets [8] [24] use regression-based evaluation over short videos with known distributional biases, and are incompatible with the multiple choice question (MCQ) evaluation paradigm that has become standard with VLM assessment. The problem is not that models answer incorrectly, but that we have no way to verify whether a correct answer reflects if the model has temporal understanding or a lucky guess. The solution is conceptually simple: every correct answer should carry a verified timestamp, so that answer accuracy and temporal localization can be jointly evaluated.

To address this methodological gap, we introduce **ChronoQA**, a benchmark that directly measures a model's temporal grounding abilities. Each question is a timestamp-anchored MCQ benchmark drawn from long-form video (10–90 minutes). ChronoQA utilizes a dual-metric evaluation protocol that distinguishes models that answer correctly from models that both answer correctly and identify accurate supporting evidence.

Unlike existing benchmarks that rely on manual annotation or focus narrowly on perceptual recall, ChronoQA enforces strict temporal grounding without temporal leakage, ensuring questions must never reference time (such as "when," "at the beginning," "in the first segment"), yet every correct answer carries precise timestamp annotations locating the evidence within the video. By utilizing multimodel blind filtering and requiring timestamp-anchored answers, we eliminate the static shortcuts prevalent in existing suites and force the model to engage with the temporal axis of the video. This design ensures that model answer selections are evaluated on what they understand, not on their ability to pattern-match temporal cues in the question's text.

The benchmark is organized around a hierarchical taxonomy of five complementary reasoning categories: (i) Action Understanding, testing whether a model can identify actions, tasks, or behaviors occurring within a specific scene or setting; (ii) Needle-in-a-Haystack Retrieval, requiring precise recall of fine-grained visual and textual details (brand names, colors, clothing, on-screen text, counts, and spatial attributes) anchored to specific people, objects, or scenes distributed across the full video; (iii) Narrative Understanding, assessing holistic comprehension of the video's overarching narrative, primary subject, target audience, or overall tone, questions answerable only by a model that has processed the video in its entirety. (iv) Causal and Cross-Segment Reasoning, demanding integration of evidence across multiple disjoint video segments to infer causation, motivation, recurring patterns, and consequential relationships; and (v) Temporal Ordering, probing a model's understanding of the sequence and chronology of events, including step ordering, immediate predecessors/successors, and transition sequences between named states;

We then evaluate several SOTA Multimodal Large Language Models (MLLMs), like Gemini 3.1 Flash [17], as well as strong open-source models like Qwen3-VL [1], and Molmo2 [5] on ChronoQA to assess their ability to temporally ground their answers in MCQ-based QA tasks.

Our contributions are:

(1) ChronoQA, a long-video MCQ benchmark that requires models to both answer correctly and localize the supporting evidence in long-form video.
(2) A grounding integrity check that exposes models answering correctly for the wrong reasons, addressing a failure mode invisible to existing benchmarks.
(3) An evaluation of several leading state-of-the-art (SOTA) MLLMs and their temporal grounding abilities.

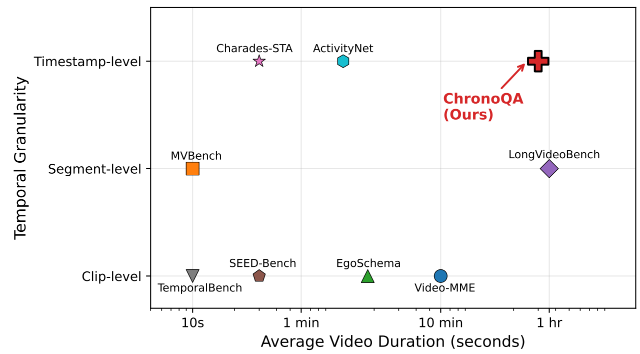

Figure 2: Positioning ChronoQA among existing video understanding benchmarks. We plot benchmarks according to their average video duration and the granularity of their temporal evaluation. ChronoQA fills a unique gap in the landscape as the only benchmark to combine long-form video (10–90 minutes) with timestamp-level temporal grounding requirements in a multiple-choice format.

2 Related Work

2.1 Video QA Benchmarks

Current Video Question Answering (VQA) benchmarks largely overlook the necessity of fine-grained temporal grounding. For example, Video-MME [7] is widely used to evaluate frontier models, spanning 900 videos, 30 subjects, and 2,700 questions. However, its temporal reasoning and event localization subtasks, among its 12 task categories, constitute a small fraction of the evaluation, and none of its questions require the model to produce or verify timestamps or time intervals. SEED-Bench [10] covers action recognition, prediction, and procedure understanding across three video dimensions, but similarly lacks any timestamp precision requirement. EgoSchema [15] tests holistic comprehension over 3-minute egocentric clips using a 5-way MCQ format and uses a temporal certificate metric to verify that correct answers require attending to the full clip rather than a single frame. However, holistic comprehension of a 3-minute clip is a different capability from timestamp-level grounding and precise localization in extended 30-90-minute durations. Although this metric captures whether holistic attention to the clip is required, it does not require the model to produce or verify a timestamp prediction. ChronoQA extends this intuition to an explicitly grounded evaluation. Rather than certifying that a question requires the full clip to answer, every question in ChronoQA carries a verified timestamp interval that the model must identify alongside its answer selection.

A separate class of benchmarks evaluates temporal structures, but at a coarse granularity. MVBench [11] treats action localization through broad beginning, middle, or end bins, effectively conflating temporal ordering with localization precision. NExT-QA [22] advances beyond descriptive benchmarks by explicitly targeting causal and temporal action reasoning in 5,440 videos, demonstrating that the best performing models achieve near-human accuracy on descriptive questions, but fail substantially on causal and temporal ones. However, NExT-QA videos average only 44 seconds in length, and no question requires the model to identify a supporting timestamp interval. Instead, the benchmark measures whether the model reasons correctly about temporal structure, not whether it can locate that structure within a longer video. Another work, TemporalBench [2], assesses sensitivity to temporal dynamics such as action frequency, motion magnitude, and event order across approximately 10K QA pairs, but does so via binary caption discrimination rather than timestamp anchoring. These benchmarks tackle temporal reasoning, understanding sequence and causality, while leaving the more rigorous problem of temporal localization largely unaddressed.

Table 1: Comparison of ChronoQA with existing video understanding benchmarks. ChronoQA is distinguished by its use of predicted timestamp intervals as a requirement for correct answering in long-form video, moving beyond the clip-level or segment-level accuracy reported by many existing suites.

Benchmark	# Videos	# Questions	Avg Duration	Timestamp	Format	Metric
Video-MME [7]	900	2,700	10 min	None	MCQ	Acc
MVBench [11]	4,000	4,000	10 sec	None	MCQ	Acc
EgoSchema [15]	5,000	5,000	3 min	None	MCQ	Acc
TemporalBench	10,000	10,000	10 sec	None	Binary	Acc
SEED-Bench [10]	20,000	19,000	30 sec	None	MCQ	Acc
NExT-QA [22]	5,440	52,000	44 sec	None	MCQ / Open-ended	Acc / WUPS
Charades-STA [8]	6,670	16,128	30 sec	Predicted	Regression	IoU
ActivityNet-QA [24]	5,800	58,000	2 min	Predicted	Regression	IoU
LongVideoBench [21]	3,500	6,678	60 min	Provided	MCQ	Acc
CG-Bench [3]	1,219	12,129	27 min	Predicted	MCQ / Open-ended	Acc / IoU / CRR
ChronoQA	**423**	**897**	**25 min**	Predicted	**MCQ**	**Acc + mIoU**

Among existing works, LongVideoBench [21] most closely aligns with the objectives of ChronoQA, featuring 6,678 MCQs for videos up to one hour. However, its "referring reasoning" format provides timestamps as input features, inverting the grounding task, prompting models to describe, or reason about, events at a given interval rather than identifying the interval corresponding to a described event.

2.2 Temporal Reasoning in Vision-Language Models

Current VLMs share a common architectural bottleneck: a vision encoder converts sampled frames into patch tokens projected into an LLM backbone, discarding inter-frame temporal signals entirely. Fine-grained motion and transition information, therefore, depends on frame sampling density, which remains sparse even in frontier models. Proprietary systems have attempted to compensate through larger context windows. Gemini 2.5 Pro [9] supports approximately six hours of video within a two-million-token context, and the natively multimodal GPT-5 [19] achieves 86.7% on Video-MME. Yet, even with these expanded budgets, VLMs exhibit systematic performance degradation as video duration increases, indicating that scaling context length alone does not resolve the underlying temporal grounding deficit [26]. Open-source models like Qwen3-VL and Molmo2 introduce spatio-temporal grounding and video tracking, but operate at smaller effective temporal resolutions.

This architectural limitation has measurable consequences for model behavior. TempCompass [14] evaluates temporal perception across five aspects: action, speed, direction, attribute change, and event order. Using conflicting video pairs that share identical static content but differ along a single temporal dimension, they neutralize single-frame shortcuts and language priors. Across eight SOTA Video LLMs, models perform near-randomly on speed, direction, and attribute change, achieving reasonable accuracy only on action, a category largely solvable from static features. NExT-QA [22] demonstrates a parallel trend in reasoning, where the disparity between model and human performance is significantly greater for causal and temporal questions than for descriptive ones, even when dealing with brief and well-structured videos. Categorical failures appear even on simpler temporal tasks. SpookyBench [20] presents information exclusively through temporal noise sequences, stripping away all spatial content, resulting in fifteen SOTA VLMs scoring 0% on tasks where humans achieve over 98% accuracy. Another work, VLM4D [25], illustrates that models frequently fail to resolve basic physical dynamics, such as distinguishing clockwise from counterclockwise rotation or correctly labeling the direction of lateral movement. Together, these results suggest that current VLMs exploit static visual shortcuts wherever available and fall back towards chance when those shortcuts are removed.

Critically, both TempCompass and NExT-QA operate on short clips where temporal evidence is local, and the search window is small. Whether this failure pattern holds in long-form video remains unanswered. Existing temporal grounding datasets, such as Charades-STA[8] and ActivityNet-QA[24], use regression-based evaluation but are similarly confined to short videos with known positional biases. Only 40% of ActivityNet moments span more than 30% of the video duration, allowing models to exploit distributional shortcuts rather than performing localization. No existing benchmark jointly measures answer accuracy and timestamp precision in a long-form setting. ChronoQA addresses this gap with a dual-metric evaluation protocol that reports both QA accuracy and mIoU over predicted intervals on videos of 10 to 90 minutes, making explicit that aggregate answer accuracy without grounding verification cannot be taken as evidence of temporal grounding ability.

3 Methods

3.1 Video Sourcing & Selection Criteria

YouTube videos provide a large-scale, real-world source of long-form content with diverse metadata and content types. We perform an initial large-scale scrape using varied search terms to collect over 90,000 candidate videos. We then filter the candidate videos based on duration (10–90 minutes), content type (instructional, narrative, procedural, sports, vlogs), licensing (Creative Commons or fair-use), and metadata availability. To manage this corpus, we

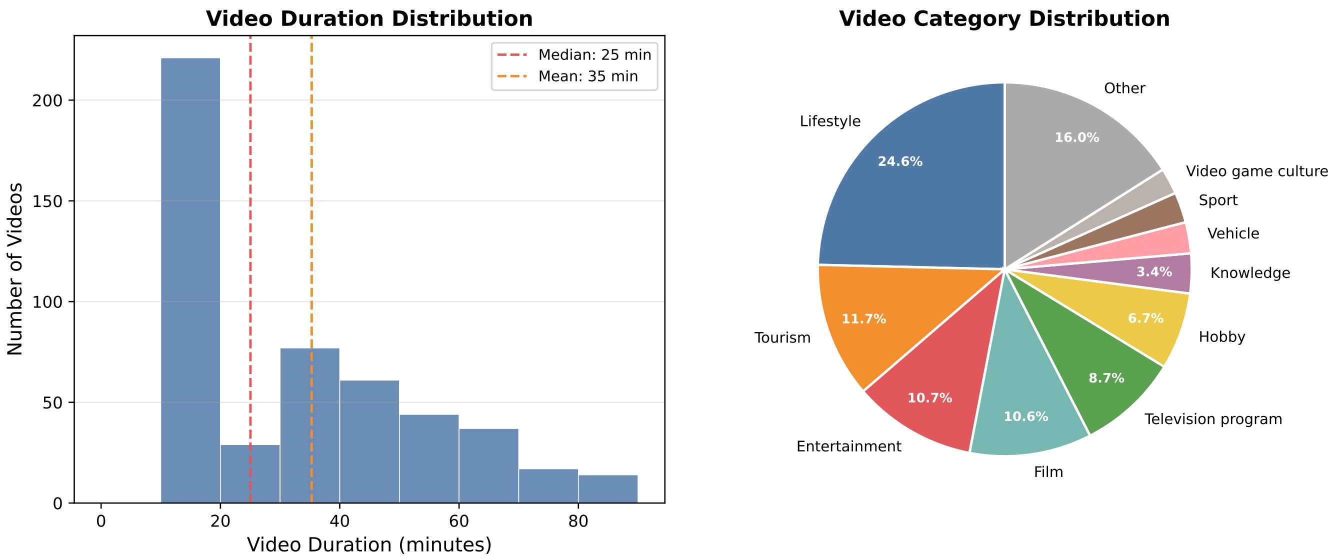

Figure 3: The benchmark is constructed from 423 long-form YouTube videos ranging from 10 to 90 minutes. The distribution is centered around a median duration of 25 minutes and a mean of 35 minutes. The dataset covers a diverse range of real-world content types to ensure broad applicability. The largest category is Lifestyle (24.6%), followed by Other (16.0%), Tourism (11.7%), Entertainment (10.7%), and Film (10.6%). Other represented niches include Hobby (8.7%), Television programs (6.7%), and Knowledge (3.4%).

categorize the filtered videos into two primary tiers based on a duration-descending sort. We create a primary collection of the 30,000 longest-duration videos (averaging 20–34 minutes), prioritizing maximum temporal depth, and then a secondary collection of 60,000 videos (averaging 20 minutes) used to ensure a broad distribution of shorter yet still long-form content. A visual diversity filter is applied to ensure sufficient temporal dynamics for meaningful temporal reasoning questions.

To ensure the dataset contains sufficient temporal dynamics for meaningful reasoning, we apply a visual diversity filter using a delta (Δ) score derived from CLIP (openai/clip-vit-base-patch32) embeddings [18]. For each video, we sample n frames at 10-second intervals (up to $n = 10$). We extract feature vectors $\mathbf{v}_i$ for each frame and normalize them to unit length such that $\hat{\mathbf{v}}_i = \mathbf{v}_i / \|\mathbf{v}_i\|_2$. The diversity score Δ is defined as the complement of the average pairwise cosine similarity across all unique frame pairs:

$$\Delta = 1 - \left(\frac{1}{n(n-1)} \sum_{i=1}^{n} \sum_{j \neq i}^{n} \hat{\mathbf{v}}_i^{\top} \hat{\mathbf{v}}_j \right)$$

This metric quantifies the visual variance within the video. A higher Δ indicates more frequent scene transitions or significant motion, whereas a low score suggests static or highly repetitive content (e.g., a static camera interview). We utilize this score to systematically exclude visually stagnant videos, ensuring the final collection requires active temporal grounding to resolve question-answer pairs. From there, we initially randomly sample 250 videos from each collection to generate our QA pairs.

3.2 QA Generation

3.2.1 Overview. We present an automated pipeline for generating temporally-grounded multiple-choice questions from equal-duration video narration captions. The pipeline operates in five stages:

(1) Caption Generation
(2) Question-Answer Generation
(3) Validation and Cleaning
(4) Distractor Synthesis
(5) Blind Filtering

All LLM inference was performed using the vLLM inference engine with Pydantic-enforced [6] structured outputs to guarantee well-formed JSON while minimizing post-hoc parsing.

3.2.2 Stage 1: Caption Generation. In order to obtain precise event timestamps, we segment each video into fixed-length clips of 10 seconds. Each segment is passed to Qwen3-VL-30B-A3B-Instruct, which samples frames at 1 frame per second, and projects them into a pixel-budget space of $P_{Video} = C \cdot 32 \cdot 32$ pixels, where C is a user-specified token budget for all sampled frames. Ten-second clips provide sufficient temporal resolution to capture fine-grained event details without over-generalizing across broader scene transitions. In our implementation, we define $C = 640$, allocating 64 tokens to capture the detail of each frame sampled at 1 FPS in a 10 second clip.

All segment inputs are submitted concurrently to the vLLM inference engine, generating one free-form caption per segment, which is recorded alongside its timestamp and source video ID.

3.2.3 Stage 2: Question-Answer Generation. The pipeline takes as input a JSON file mapping video IDs to the recorded caption data, where each entry contains a list of timestamped captions and the original video duration. We define five specialized prompt templates, each targeting a distinct facet of video understanding: Action Understanding, Needle-in-the-Haystack, Narrative Understanding, Causal Understanding, and Ordering. Each template generates 3 questions per video. The model uses the captions and their timestamps to produce question-answer pairs grounded in the video content. All templates constrain questions to avoid direct temporal references (no "when," "beginning," or timestamp references), and only the correct answer carries a temporal annotation in the form of start and end timestamps for the relevant segment.

We use Qwen3-Next-80B-A3B-Thinking [23] constrained to a Pydantic schema via vLLM's `StructuredOutputsParams` to guarantee each response contains well-formed question-answer pairs with associated timestamps. QA generation succeeds approximately 98% of the time, with failures arising from output budget overruns or malformed JSON responses.

3.2.4 Stage 3: Validation and Cleaning. The generated QA pairs then undergo three quality control steps.

Following generation, a pattern-matching filter removes any temporally self-referential questions, meaning questions whose answer is a timestamp or segment reference (e.g., "When was X shown?", "In which segment did Y occur?"). Such questions are structurally incompatible with ChronoQA's evaluation design because the benchmark measures a model's ability to produce a content answer and a grounding interval separately. Questions that collapse these two into a single temporal response cannot be scored in any principled way. Although the generation prompt is explicitly conditioned against producing such questions, this filter serves as a deterministic safeguard to ensure none reach the final dataset.

Next, we apply timestamp merging to consolidate multiple timestamp entries, merging adjacent segments separated by less than 2 seconds into a single continuous range to avoid fragmentation. Finally, a timestamp jitter is applied to increase ground-truth variability. Because captions are generated from fixed 10-second clips, LLM-generated timestamps are ordinarily constrained to 10-second boundaries. To produce more varied annotations, the recorded start timestamp is shifted earlier by $X \sim \mathrm{Unif}(\{1, 2, 3\})$ seconds ($X \sim \mathrm{Uniform}(1, 3)$) and the end timestamp later by $X \sim \mathrm{Unif}(\{1, 2, 3\})$ seconds, stopping at the existing video's boundaries (Can't be less than 0, or longer than the video). This also provides tolerance for imprecise annotations when ground-truth ranges are smaller than the clip length.

3.2.5 Stage 4: Distractor Generation. For each question, four plausible but incorrect answer options (distractors) are generated in a separate LLM pass. The model is instructed to produce options that (a) are plausible within the question context, (b) belong to the same semantic category as the correct answer, (c) match the correct answer in length and casing, and (d) are unambiguously incorrect. When relevant caption data is available, narration segments within a ±30-second margin around the correct answer's timestamp are extracted and provided as localized context to ground distractors in actual video content. Questions for which fewer than four valid distractors are generated are discarded, and cannot be part of the final set. The correct answer and four distractors are randomly shuffled, and the position of the correct answer is recorded as a letter (A-E).

3.2.6 Stage 5: Blind Filtering. Language models can sometimes identify the correct answer without visual input by exploiting shared linguistic patterns between the question and the ground truth, particularly when distractors are lightly edited variants of the correct answer [2]. To remove questions susceptible to this bias, we apply a blind filtering procedure in which each QA sample, consisting only of the question text and five answer options, with no videos, is presented to five SOTA LLMs. Each model is prompted to select the correct answer under the same task framing used in the full evaluation. Any sample for which at least two models selects the correct answer is considered language-solvable, and is removed from the dataset, ensuring that retained questions require visual grounding rather than linguistic pattern-matching. The text-only solvability rate of the retained set is constrained to at or below the random baseline for a five-way multiple-choice question at 20%.

3.2.7 Scale Statistics. Each final QA instance contains a video ID, question text, the correct answer letter, and five answer options (each with associated timestamps), formatted as structured JSON suitable for use in video understanding benchmarks.

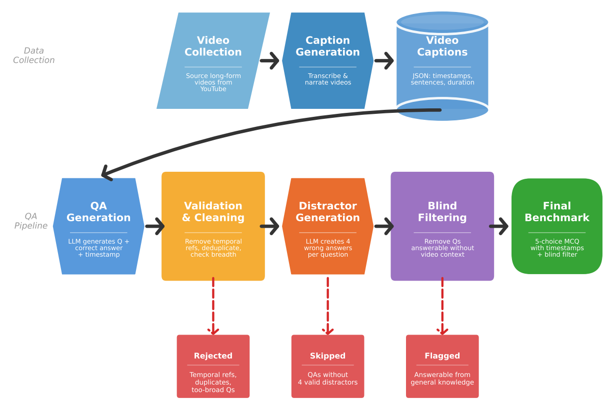

Figure 4: A visualization of ChronoQA's generation pipeline. Specialized prompt templates generate questions based on 10-second video segment captions, specifically avoiding temporal language such as "when" or "at the beginning". Then we pattern-match to discard questions that require timestamps to solve, as well as timestamp merging and jittering to increase ground-truth variability. Next, for every question, four plausible but unambiguously incorrect distractors are generated that belong to the same semantic category as the correct answer. Finally, to ensure the benchmark requires visual grounding, questions are presented to SOTA LLMs without the video. Any question solvable by linguistic pattern-matching alone is removed.

3.2.8 Human Baseline Collection. To establish a human performance reference point, we collect annotations from a single evaluator. The evaluator is provided access to the full source video for each instance alongside the question and five answer options, and is instructed to select the correct answer and to annotate the specific timestamp interval within the video they believe best supports that answer, which is the same dual-response format used in model evaluation. No time constraints are imposed. The evaluator completed 100 instances drawn from a stratified random sample across the five task categories. Human QA accuracy on this set was 91.2%, and human mIoU was 82.5%, establishing a performance ceiling substantially above all evaluated models on both metrics. We acknowledge that single-annotator baselines do not permit the calculation of inter-annotator agreement and may reflect individual variation in timestamp granularity. We therefore present these figures as an indicative upper bound rather than a definitive measure of human capability on ChronoQA, and identify the collection of a multi-annotator human baseline as a priority for future work.

4 Benchmark Design

4.1 Task Formulation

Given a long-form video V and a natural language question Q that references a temporal event, transition, or property of the video, the model must select the correct answer a_i from a set of **five** options $\{a_1, \ldots, a_5\}$. Each option consists of a **textual description** indicating the segment of the video the description refers to.

This task requires the model to:

(1) Interpret the contextual reference in Q to identify the relevant moment(s) of the video.
(2) Inspect the corresponding video segment(s).
(3) Select the answer whose textual description correctly matches the visual content of that segment.

Incorrect options are wrong because their descriptions are visually false **at that particular moment in time**. A model without this design ensures that contrastive reasoning of answer options must be done to eliminate distractors. A model without visual context has no basis for discrimination beyond linguistic guessing, achieving near-random performance (20%) on well-constructed questions.

4.2 Task Taxonomy

ChronoQA organizes its questions into five categories, each targeting a distinct capability. These categories are roughly ordered by the scope of video context required, from segment-based visual retrieval to full-video narrative comprehension.

4.2.1 Action Understanding. **Goal:** Identify what action is occurring at a specific moment in the video, including procedural details on how the tasks or activities are being carried out.

Action Understanding questions require the model to accurately characterize a segment's content and how an action unfolds, not merely that it is occurring. This demands fine-grained visual perception of tool usage, body posture, object manipulation, and sequential steps within a single continuous action. Questions in this category are anchored to a specific segment and do not require cross-segment reasoning.

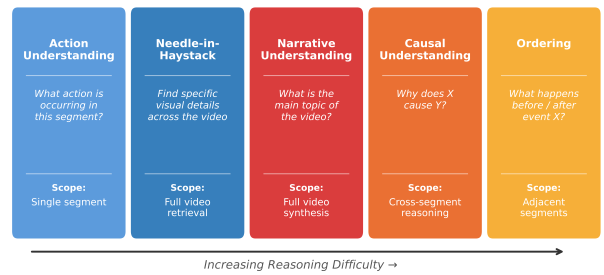

Figure 5: ChronoQA's Task Taxonomy. Questions are organized into five categories of increasing reasoning difficulty: Action Understanding (local), Needle-in-the-Haystack (retrieval), Narrative Understanding (global), Causal Understanding (cross-segment), and Ordering (sequential). Each category tests a distinct capability, from fine-grained perception of procedural details to the synthesis of an overarching narrative arc.

What it addresses: Existing benchmarks like MVBench [11] test coarse action recognition (e.g., "is the person running or walking?") but rarely probe procedural details. **ChronoQA**'s Action Understanding questions demand the level of detail a domain expert would notice. For example, identifying the difference between two similar techniques performed at different moments in a tutorial video.

4.2.2 Needle-in-the-Haystack. **Goal:** Retrieve a specific visual or textual detail, like an object's characteristics, brand names, colors, materials, on-screen text, or object counts, from anywhere within the full video timeline.

Needle-in-the-Haystack questions test a model's ability to locate and report precise details that only appear briefly on-screen. Correct-answer timestamps are uniformly distributed across the full video duration, meaning target details are equally likely to appear early, in the middle, or near the end. This design counters the start/end positional bias documented in prior temporal grounding datasets, where segments cluster near video boundaries, enabling models to exploit positional priors. The model must sustain attention across the entire video and cannot rely on narrative context to narrow the search window.

What it addresses: Most video benchmarks either test contextually prominent events (that are therefore easy to anticipate) or sample questions from predictable temporal positions. Needle-in-the-Haystack isolates retrieval capabilities from reasoning capabilities, creating a pure test of a model's ability to find a specific detail in a long, unstructured visual stream.

4.2.3 Narrative Understanding. **Goal:** Assess high-level synthesis of a video's overarching narrative arc, primary subject, core message, or overall mood and tone. This requires the model to have processed the video in its entirety.

Narrative Understanding questions test whether a model has formed a coherent global understanding of the video, rather than a collection of local observations. Correct answers require understanding a video's broader context, including its genre, overarching goal, main thesis, primary technique, or dominant tone. Questions must be anchored to properties that cannot be determined from any single representative segment. Despite testing high-level understanding, all questions are held to strict objectivity standards, and questions without a single defensible correct answer are excluded during filtering.

What it addresses: SEED-Bench includes "action prediction" and "procedure understanding" as high-level video tasks, but these tasks focus on fine-grained categories of basic actions and action segmentation. Video-MME [7] includes "summarization" as a task category, but answers are often derivable from a few key scenes. **ChronoQA**'s Narrative category specifically enforces full-video dependency. The correct answer should reflect the *aggregate* of the video's content, not any one segment of the video.

4.2.4 Causal Understanding. **Goal:** Integrating evidence from two or more non-adjacent segments to infer motivations, cause-and-effect relationships, or patterns that only become apparent across multiple scenes.

Every Causal Understanding question is constructed such that **no single video segment contains sufficient information to answer correctly**. The model must synthesize observations from multiple temporally separate parts of the video. This may involve understanding a character's motivation for a particular action, identifying an earlier event that caused a later outcome, or recognizing a recurring pattern whose significance emerges through repetition. This category establishes a distinction between comprehension (understanding what happens at a moment) and reasoning (understanding why, given what came before and after). It is also the category most resistant to frame-sampling shortcuts. A model that processes only a subset of frames may observe an effect without seeing its cause.

What it addresses: EgoSchema [15] requires holistic reasoning across 3-minute clips, but its questions are answerable from the general arc of the clip. LongVideoBench [21] tests referring reasoning, but supplies relevant timestamps in the question. Neither requires the model to independently identify and integrate evidence from multiple non-adjacent segments. This category makes cross-segment synthesis a structural requirement.

4.2.5 Ordering. **Goal:** Reason about the sequential structure of events and actions in the video, such as what happens before or after a specific event, how transitions occur, and in what order steps are taken.

Ordering questions probe the model's ability to temporally sequence events. The model must understand what occurs at a given moment, and how that moment relates to adjacent events. This includes direct before/after relationships, scene transitions, and step ordering within processes. Unlike Action Understanding, the correctness depends on the position of an event relative to others, not solely on local content. Unlike Causal Understanding, no inference about why events occur in a particular order is required.

What it addresses: MVBench [11] includes sequencing-based tasks but primarily uses short clips where temporal order can often be inferred from a handful of frames. **ChronoQA**'s ordering questions span long-form videos where events may be separated by spans of several minutes, requiring the model to maintain a coherent temporal mapping of the video rather than simply relying on local context.

4.3 Design Principles

ChronoQA is designed around four core principles that jointly ensure benchmark integrity. First, questions require reasoning over events separated by minutes rather than seconds, preventing single-frame shortcuts. Second, correct answers are uniformly distributed across positions $A-E$ to eliminate positional bias. Third, ChronoQA adopts the standard five-option MCQ format, using accuracy as a primary metric while encoding temporal precision through a timestamp paired with the correct answer. This replaces the IoU-based regression metrics of prior works with a format compatible with standardized VLM evaluation. Finally, distractors are not randomly sampled,and are semantically plausible events drawn from different moments in the same video, ensuring that the task requires temporal grounding rather than event recognition.

5 Evaluation Protocol

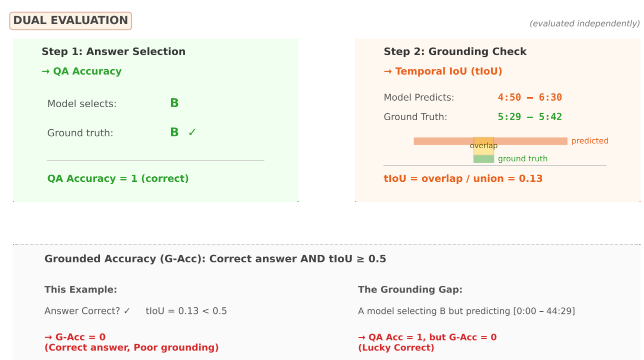

Figure 6: Illustration of a ChronoQA task and the dual-metric evaluation protocol. Models are provided with a long-form video and a question that avoids direct temporal references. To succeed, a model must select the correct textual description (Step 1) and independently predict the supporting timestamp interval (Step 2). The grounding gap is revealed when a model achieves high QA accuracy but low temporal overlap (tIoU), indicating a "lucky correct" answer based on linguistic priors rather than visual evidence.

5.1 Evaluated Models

ChronoQA evaluates four models spanning the full range of capabilities in the current VLM landscape, including both closed-source and open-source models.

All proprietary models are evaluated through their respective public APIs at the most capable available version at the time of paper writing. Open-source models are evaluated using their official HuggingFace checkpoints at their default recommended inference configuration.

Table 2: Models evaluated on ChronoQA.

Model	Provider	Access	Context / Frame Limit
GPT-5.4 [19]	OpenAI	API	400K tokens
Gemini 3.1 Flash [17]	Google	API	Native long-video; up to 1M tokens
Qwen3-VL-8B [1]	Alibaba / Open-source	Local Inference	256K (dynamic resolution, expandable to 1M)
Molmo2-8B [5]	Allen AI / Open-source	Local Inference	36K

5.2 Input Specification

We provide each model with a standardized input consisting of three components:

(1) **Video Input:** The source video, formatted as uniformly sampled frames at a rate standardized to a practical input limit for the model. For models with native long-video ingestion (e.g., Gemini), the full video is passed directly. For all other models, uniform frame sampling is applied.
(2) **Question Text:** The natural language question Q.
(3) **Answer Options:** Five labeled options (A-E), each consisting of a textual description and its associated timestamp interval $[t_{start}, t_{end}]$.

The full prompt template is fixed across all models. No chain-of-thought instructions are included in the primary evaluation (zero-shot direct answer). To ensure fair comparison between models with different input limits, we defined a standardized frame budget of 64 uniformly sampled frames as the default evaluation setting.

5.3 Metrics

ChronoQA employs two complementary evaluation metrics targeting distinct aspects of a model's capabilities. One measures the accuracy of the final answer selection, and the other measures the quality of temporal grounding that underlies correct answers. This dual-metric design is directly motivated by [3], which demonstrated that MCQ accuracy alone is insufficient to verify that correct answers are genuinely grounded. A model may select the right answer through surface-level reasoning without ever identifying the relevant video segment.

5.3.1 QA Accuracy (Primary Metric). QA Accuracy is the percentage of questions for which the model selects the correct answer option:

$$\text{Acc} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\hat{a}_i = a_i^*] \tag{1}$$

where N is the total number of questions, $\hat{a}_i$ is the model's selected option, and a_i^* is the ground-truth correct option.

QA Accuracy is reported at two levels of granularity:

- **Overall Accuracy**: Computed across all questions in the dataset.
- **Per-Category Accuracy**: Separately for each of the five task categories (Action Understanding, Needle-in-the-Haystack, Narrative Understanding, Causal Understanding, and Ordering).

The random baseline for a 5-option MCQ is 20%. A text-only baseline (no video input) must achieve below 20% to confirm that linguistic patterns in the question and answer text do not disambiguate the correct option.

5.3.2 Mean Temporal IoU (mIoU) - Grounding Metric. mIoU measures the quality of a model's temporal localization. Given a question, how accurately does the model identify the relevant video segment, regardless of whether it selects the correct answer? Following the standard temporal grounding evaluation protocol, temporal IoU (tIoU) for a single prediction is defined as:

$$\text{tIoU} = \frac{\sum_{i\in\mathcal{G},\, j\in\mathcal{P}} \max\left(0, \min(b_i,d_j)-\max(a_i,c_j)\right)}{\sum_{i\in\mathcal{G}}(b_i-a_i)+\sum_{j\in\mathcal{P}}(d_j-c_j)-\sum_{i\in\mathcal{G},\, j\in\mathcal{P}} \max\left(0, \min(b_i,d_j)-\max(a_i,c_j)\right)} \times 100\% \tag{2}$$

where a_i, b_i are the start and end timestamps of the i-th ground truth interval of G. c_j, d_j are the start and end timestamps of the j-th predicted interval of P. We calculate the mean IoU (mIoU) by averaging the tIoU scores obtained by the model across all question queries.

$$\text{mIoU} = \frac{1}{N} \sum_{i=1}^{N} \text{tIoU}_i \tag{3}$$

Eliciting Timestamp Predictions. Since ChronoQA is an MCQ benchmark and most evaluated models are not natively trained for timestamp regression, timestamp predictions are obtained with a structured prompting strategy. After selecting an answer option, models are further prompted to output the specific timestamp interval within the video that they believe best supports their answer (e.g., "State the start and end time in seconds of the video segment that directly answers this question."). This protocol follows the white-box evaluation paradigm of CG-Bench [3], which requires models to explicitly output grounding intervals alongside their MCQ selections.

For models that refuse, or fail, to produce a parsable timestamp (e.g., outputting a range covering the full video or a null response), a fallback interval of $[0, T]$ (full video duration) is assigned, yielding a tIoU score proportional to the fraction of the video covered by the ground-truth interval. This penalizes uninformative outputs without zeroing them entirely.

mIoU as a grounding integrity check. mIoU serves a diagnostic function beyond ranking, allowing us to detect cases where a model selects the right option but provides a grounding interval inconsistent with the correct timestamp, or a "lucky correct" answer. A model with high QA Accuracy but low mIoU is selecting correct answers for the wrong reasons, which CG-Bench [3] was designed to surface. We additionally report Grounded Accuracy ($G - Acc$), which is the percentage of questions where the model both selects the correct answer and achieves $tIoU \geq 0.5$ on the grounding interval, providing a joint measure of answer correctness and temporal grounding quality.

Recall@IoU (Supporting Metric). Following CG-Bench, in order to improve the robustness of question grounding evaluation, we additionally report Recall@IoU, measuring the probability of successfully recalling clue intervals at various IoU thresholds. We calculate the average recall rate at IoU thresholds of 0.1, 0.2, 0.3, 0.4, and 0.5 to determine the final result. This metric captures the distribution of grounding quality across the full IoU spectrum, providing a richer picture than mIoU alone, particularly for distinguishing models that frequently achieve partial overlap (IoU 0.3) from those that more often achieve precise localization (IoU 0.5+).

5.4 Answer Extraction

Reliable answer extraction is essential for consistent scoring across models with varying output styles. We adopt a two-stage extraction procedure, following MMBench [13]:

Rule-based extraction (primary): Parse the model's output for an explicit option label (A, B, C, D, or E) using regex matching. This succeeds for the majority of well-behaved model outputs. LLM-based extraction (fallback): For outputs that do not contain an unambiguous option label, a separate GPT-4o [16] judge is prompted with the model's full response and the original answer options to identify the intended selection. This handles cases where a model re-states the answer text rather than the option letter.

Timestamp extraction follows a similar two-stage approach, starting with regex parsing for structured outputs (e.g., "Start: 34.5s, End: 42.1s"), then followed by an LLM-based parser for free-form outputs. If a model explicitly refuses to answer, or if both extraction stages fail, the question is scored as incorrect (0 accuracy) and assigned a full-video fallback interval for tIoU computation. Refusal rates are reported per model as a diagnostic.

5.5 Baseline Configurations

Four baseline conditions are evaluated to establish lower bounds and verify benchmark integrity. All baselines use the same answer extraction pipeline.

The text-only baseline is the most critical integrity check. If any text-only model exceeds 20% accuracy overall, the affected questions are flagged for manual review and potential removal from the evaluation set, following MMStar's [4] multi-modal gain filtering procedure.

5.6 Reproducibility and Access

All evaluation code, prompt templates, answer extraction scripts, and model inference configurations are released publicly.

6 Experiments & Results

6.1 Main Results

In Table 3, we report the performance of four state-of-the-art video-language models on our ChronoQA benchmark. Overall, none of the models achieve a G-Acc (Grounded Accuracy) exceeding 5.16%. Among the instances evaluated, Molmo2 achieves the highest scores across all shared question categories, notably reaching 80.7% accuracy on Narrative Understanding and 73.2% on Causal Understanding. These results challenge the assumption that parameter scale is the primary driver of performance on temporally demanding video

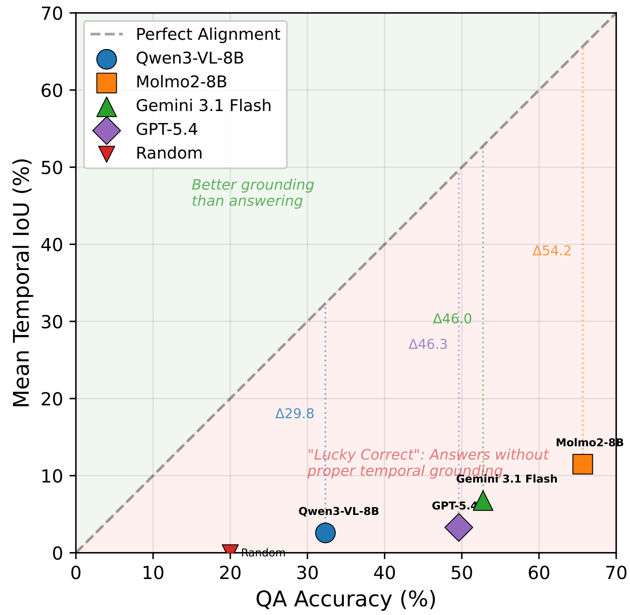

Figure 7: The Grounding Gap: QA Accuracy vs. Mean Temporal IoU (mIoU). Most evaluated models achieve high answer accuracy while failing to localize the supporting evidence. The persistent gap between answer correctness and grounding precision suggests that models often exploit surface-level shortcuts or common-sense heuristics rather than performing temporal reasoning.

tasks. Within the evaluated subset, Molmo2's performance suggests that training efficiency and video-native architectural design may matter more than model size for this class of task, a finding that warrants further investigation at full benchmark scale.

Despite the high QA accuracy typically associated with models like GPT-5.4, there remains a profound grounding gap where models succeed in identifying what happened but fail to localize *when* it occurred. The grounding gap underlying this result is examined in detail in Section 6.2.

6.2 The Grounding Gap

Molmo2 achieves the highest mIoU of any model at 11.48%, compared to 8.0% for Gemini 3.1 Flash and 3.3% for GPT-5.4. This pattern is consistent with the hypothesis that specialized open-source models may be better suited to high-density temporal reasoning than general-purpose systems trained primarily for language. A recurring phenomenon is observable across all models regardless of scale. GPT-5.4, for instance, achieves a robust 49.6% QA accuracy on its full evaluated set yet produces an mIoU of just 3.3%, a gap that is difficult to attribute to dataset artifacts given ChronoQA's blind filtering controls. Qwen3-VL-8B shows a similar pattern at 32.3% accuracy and 2.6% mIoU.

These scores indicate that models are exploiting surface-level cues like answer plausibility, common-sense elimination, and linguistic coherence, rather than engaging in temporal reasoning. The

Table 3: We report QA Accuracy, Mean Temporal IoU (mIoU), and Grounded Accuracy (G-Acc), where a successful trial requires both the correct answer and a $tIoU \geq 0.5$. Human performance significantly outperforms all models, particularly in grounding precision (mIoU), where even the best-performing model, Molmo2, falls below 11%.

Model	QA Acc	mIoU	G-Acc	Recall@0.1	Recall@0.5
GPT-5.4	0.4961	0.0329	0.0078	0.1081	0.0100
Qwen3-VL-8B	0.3233	0.0258	0.0067	0.0502	0.0145
Gemini 3.1 Flash	0.6496	0.0796	0.0197	0.1929	0.0394
Molmo2-8B	0.6566	0.1148	0.0516	0.2683	0.0800
Human	**0.9124**	**0.8245**	**0.7631**	**0.9218**	**0.7829**

disconnect is most visible in Narrative Understanding, where both Molmo2 and GPT-5.4 exceed 80% QA accuracy while their mIoU scores remain below 11%, underscoring that high answer-level performance does not imply evidence-level grounding.

6.3 Baseline Integrity Results

To assess the degree to which models exploit dataset shortcuts rather than performing genuine temporal reasoning, we conducted a blind filtering evaluation in which answer choices were presented without any accompanying video or temporal context. A model that performs substantially above chance under these conditions can be presumed to rely on linguistic priors or common-sense heuristics rather than video comprehension. Across all models, blind accuracy remained near or below the expected random baseline for the multiple-choice format, confirming that ChronoQA questions resist shortcut exploitation and require video-grounded reasoning. This result validates the integrity of our benchmark and strengthens the interpretation that the grounding gap documented in Table 3 reflects true model deficiencies rather than artifacts of question design.

6.4 Performance by Video Duration

To examine how video length affects model performance, we stratify results across four duration bins: short (≤20 minutes), medium (20–40 minutes), long (40–60 minutes), and extended (>60 minutes), shown in Table 4. Across evaluated models, performance metrics generally exhibit a downward trajectory as video duration increases, reinforcing the hypothesis that extended temporal contexts impose significant demands on long-range reasoning and evidence retrieval. Molmo2-8B demonstrates the most robust grounding performance, maintaining a superior mIoU compared to its peers across all categories, though it experiences a monotonic decline from 14.3% in the short bin to 4.1% in the extended bin. Conversely, GPT-5.4 and Qwen3-VL-8B struggle to maintain grounding precision, with mIoU often remaining below 4% and 5%, respectively, in the longer bins, suggesting that fixed-rate frame-sampling strategies may become inadequate as temporal density increases. Gemini 3.1 Flash shows a unique performance profile across the four duration bins. While its QA accuracy remains resilient (recovering to 66.7% in the extended bin), its grounding capability is highly volatile, peaking at 14.6% mIoU in the long bin before regressing to 3.2% for videos exceeding 60 minutes. These findings underscore that temporal scalability, specifically the ability to maintain precise grounding in high-density contexts, remains a critical bottleneck for the development of next-generation video-language models.

6.5 Performance by Task Category

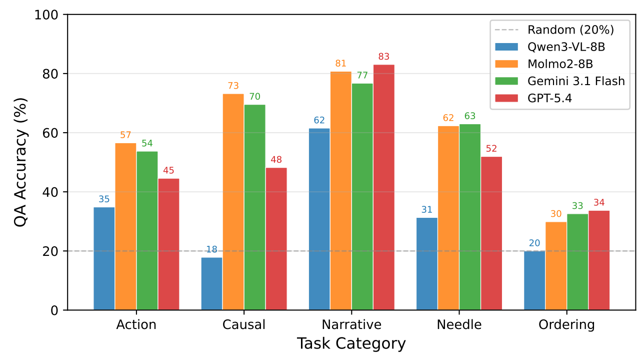

Figure 8: Breakdown of model performance across task categories. While models perform best on Narrative Understanding, which often admits linguistic inference, they struggle significantly with Temporal Ordering and Causal Reasoning. Molmo2 consistently demonstrates superior performance in Needle-in-the-Haystack tasks, suggesting that its architecture is better suited for dense temporal localization than fixed-rate frame-sampling strategies.

Figure 8 depicts the QA accuracy and mIoU across the five task categories in ChronoQA. Performance varies substantially across categories, revealing that each poses distinct challenges to current video-language models.

Narrative Understanding is the category in which models perform best overall, with Molmo2 and GPT-5.4 reaching 80.8% and 83.1% QA accuracy, respectively. This is unsurprising given that narrative-level questions often admit linguistic inference, and models may recover correct answers from partial video comprehension. Causal Reasoning yields moderate QA accuracy across models, with Molmo2 leading at 73.2%, but near-zero grounding performance, indicating that models arrive at correct causal conclusions through plausibility rather than by identifying the precise temporal evidence that supports them.

Action Understanding presents a more uniform challenge. Molmo2 again leads at 56.6% QA accuracy and 12.5% mIoU, while GPT-5.4

Table 4: Model Performance (QA Accuracy / mIoU) Across Video Duration Bins

Model	≤20 min	20–40 min	40–60 min	>60 min
Molmo2-8B	70.0% / 0.143	66.7% / 0.095	67.0% / 0.067	63.9% / 0.041
Gemini 3.1 Flash	66.2% / 0.081	66.7% / 0.056	54.8% / 0.146	66.7% / 0.032
GPT-5.4	53.0% / 0.039	45.9% / 0.022	46.9% / 0.032	48.6% / 0.032
Qwen3-VL-8B	32.3% / 0.017	28.5% / 0.025	40.1% / 0.050	27.0% / 0.021

(44.6%) and Qwen3-VL-8B (34.9%) trail considerably. The NIAH category, designed to probe fine-grained moment retrieval within long videos, exposes the sharpest performance disparities. Molmo2 achieves 62.34% QA accuracy and 14.9% mIoU on this task, substantially outperforming GPT-5.4 (52.0% accuracy, 1.9% mIoU) and Qwen3-VL-8B (31.3% accuracy, 2.5% mIoU), confirming that frame-sampled models are ill-suited for tasks requiring dense temporal localization.

Temporal Order is the most challenging category across the board, with all models falling below 34% QA accuracy and below 2.4% mIoU. This suggests that understanding the sequential structure of events within a video remains an open problem for all model architectures evaluated.

7 Analysis & Insights

7.1 Error Taxonomy and Qualitative Analysis

Qualitative inspection of model outputs reveals three recurring failure patterns that account for the majority of errors across models and categories.

The first and most prevalent failure mode is *temporal displacement*, in which a model identifies the correct event or action but attributes it to the wrong temporal segment. This error is particularly common in NIAH and Temporal Order tasks, where multiple superficially similar events occur across a video, and the model lacks the resolution to discriminate among them. GPT-5.4 and Qwen3-VL-8B exhibit this pattern most frequently, consistent with their reliance on sparse frame sampling.

The second failure mode is *plausibility substitution*, where a model selects an answer that is semantically coherent and contextually reasonable but not grounded in the actual video content. This error is most evident in Causal Reasoning tasks, where questions of the form "why did X happen" admit plausible answers that require no temporal evidence. The near-zero grounding scores in this category across all models are consistent with a pattern in which correct QA answers are reached through inference rather than observation. Models may draw on general world knowledge or narrative plausibility rather than attending to specific video content. Confirming this mechanism directly would require analysis of model reasoning traces, but the grounding evidence alone suggests that causal questions are particularly susceptible to this form of error.

The third pattern is *boundary imprecision*, in which a model correctly identifies the relevant temporal region but produces grounding intervals that are too broad or misaligned relative to the ground-truth annotation. This manifests as low Recall@τ at stricter IoU thresholds even when coarse localization succeeds, and is visible across all models in the rapid drop-off between Recall@0.1 and Recall@0.5. Molmo2 is least affected by this pattern, suggesting that its training encourages tighter interval estimation.

Together, these failure modes indicate that improving temporal grounding will require not only richer video representations but also training objectives that explicitly reward precise evidence localization rather than answer correctness alone.

8 Conclusion

We introduced ChronoQA, a benchmark designed to evaluate the temporal reasoning and grounding capabilities of video-language models across five task categories: Action Understanding, Causal Reasoning, Narrative Understanding, Needle-in-a-Haystack retrieval, and Temporal Order. Unlike prior benchmarks that assess video comprehension primarily through answer-level accuracy, ChronoQA couples each question with a required temporal grounding annotation, enabling a joint evaluation of *what* a model understands and *when* it localizes supporting evidence.

Our evaluation of four representative models reveals several key findings. First, no model exceeds the overall accuracy of QA 68%, and the temporal grounding performance remains low across the board, with mIoU scores consistently below 11%. Second, a persistent grounding gap exists across all architectures, where models frequently identify correct answers while failing to localize the temporal evidence that justifies them. Third, and notably, the open-source model Molmo2 outperforms larger proprietary systems on both QA accuracy and grounding metrics, suggesting that training efficiency and video-native architecture design matter more than parameter scale for this class of tasks.

These results carry several implications for future work. Temporal grounding should be treated as a first-class evaluation criterion rather than a secondary annotation, and training objectives should explicitly reward precise interval prediction. The failure of frame-sampled models on NIAH and Ordering tasks suggests that dense or continuous video representations may be necessary for tasks involving fine-grained temporal discrimination. Finally, the consistent failure across models on Causal Reasoning grounding indicates that the field lacks adequate mechanisms for connecting causal inference to observable temporal evidence.

ChronoQA is available for continued evaluation, and we anticipate that it will serve as a useful diagnostic tool as the video-language modeling community develops architectures better suited to the demands of temporal reasoning.

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